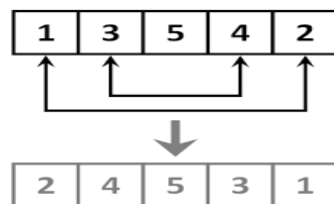


Arrays

- 1) Write a C program to input 10 numbers through the keyboard into an array and display the results of addition of even numbers and product of odd numbers.
- 2) Write a C program to input 10 numbers through the keyboard into an array and find the biggest and smallest number in an Unsorted array without using any Sorting Technique.
- 3) Write a C program to input 10 numbers through the keyboard and find the number of prime numbers count, store them into a separate array and display it.
- 4) Write a C program to find out second largest and second smallest elements of an unsorted array without using any Sorting Technique.
- 5) Write a C program to reverse the elements of a given array.



- 6) Write a C program to delete an element at desired position from an array.

1	14
2	50
3	73
4	9
5	24
6	3
7	92
8	-3

Original Array

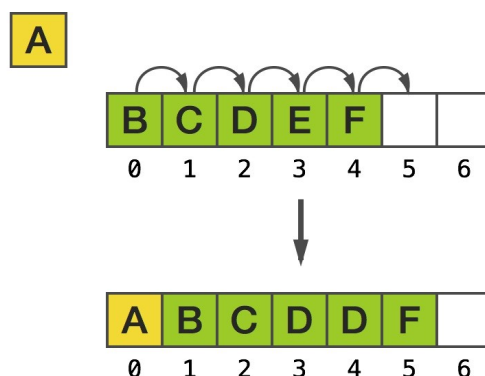
1	14
2	50
3	73
4	9
5	
6	3
7	92
8	-3

5th Element deleted – leaving an empty location

1	14
2	50
3	73
4	9
5	3
6	92
7	-3
8	

Array after Deletion

- 7) Write a C program to insert an element at desired position in an array.



For Example if 'A' is to be stored at '0' position then,

8) Write a C program which deletes the duplicate elements of an array.

Original: [A, C, B, D, A, B, E, D, B, C]

Remove duplicate result: D, E, A, B, C,

9)

Write a C program to find the duplicate elements of a given array and find the count of duplicated elements.

Ex: if `int a[] = {0,3,1,0,5,1,2,0,4,5}`

output : -

The duplicate elements are existed in an array

0 -- 3 times

1 -- 2 times

5 -- 2 times

10) Write a program to print the non repeated numbers of a given array.

Ex : if `int a[] = {0,3,1,0,5,1,2,0,4,5}`

Output : 3, 2, 4

11) Write a program to copy the elements of one array into another array without duplicate items as a first slot, and store duplicate elements as a second slot.

Ex: source array `{10,2,4,5,2,1,3,4,6,5,8,9,2}`

destination arrays `{10,2,4,5,1,3,6,8,9}` , `{ 2,2,4,5}`
 first slot second slot

Take two different arrays for first and second slots.

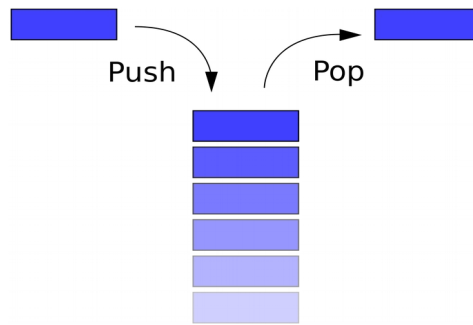
12) Write a C program to evaluate the following series. The series contains sum of square of numbers from 1 to 'n'. Store result of each term in an array. Calculate value of 'S' using array.

$$S = 1^2 + 2^2 + 3^2 + 4^2 + \dots + n^2$$
$$= [1, 4, 9, 16, \dots, n^2]$$

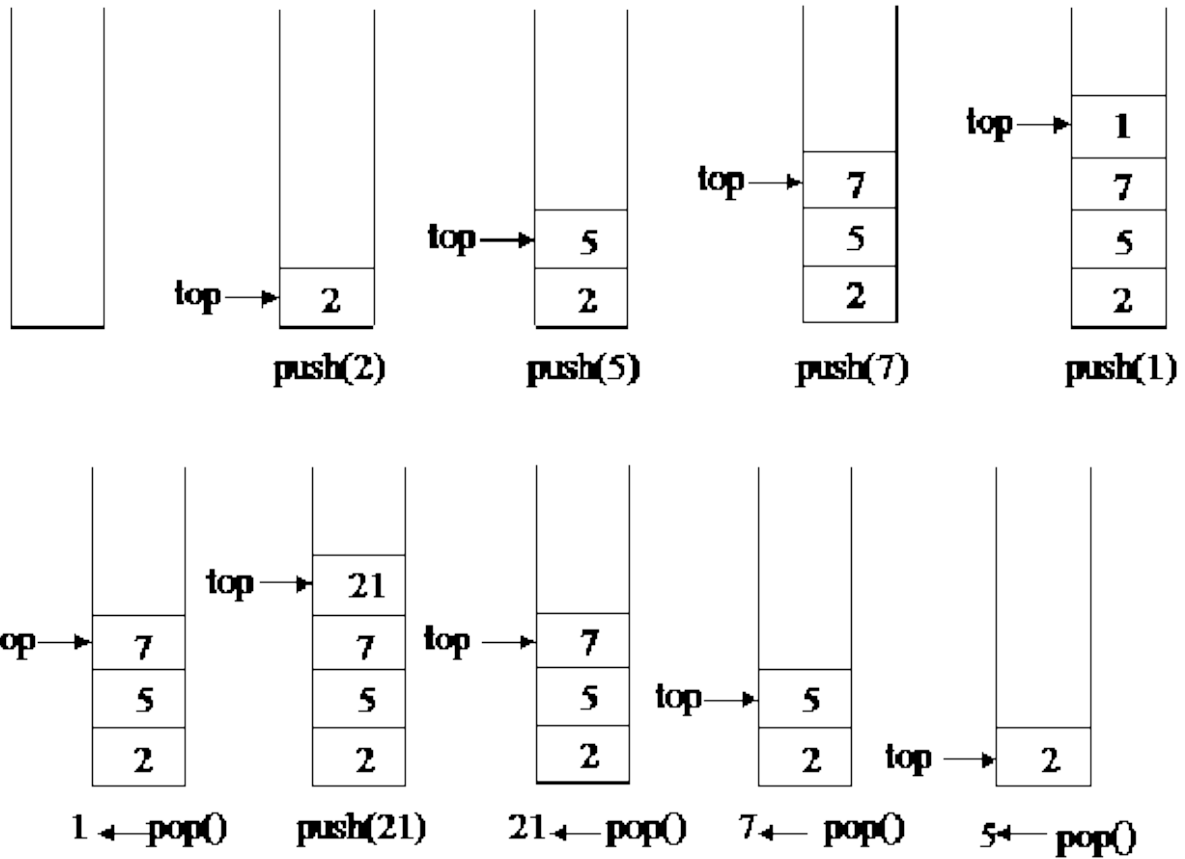
Suppose $n = 4$,

then $S = 1^2 + 2^2 + 3^2 + 4^2$;
 $S = 1 + 4 + 9 + 16$;
 $S = 30$.

13) Write a C program to implement the stack using arrays.



STACK



-----END-----